

ENHANCING NEW HIRE ONBOARDING THROUGH
QUESTION ANSWERING WITH RETRIEVAL-AUGMENTED
GENERATION TECHNIQUES

by

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Declaration

I hereby declare that this project report is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in this report.

This report has also not been submitted for any degree in any university previously.

A handwritten signature in black ink, reading "Venessa", positioned above a horizontal line.

Venessa Tan

5 September 2024

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Abstract

Enhancing New Hire Onboarding through Question Answering with Retrieval-Augmented Generation Techniques

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This thesis explores the integration of generative AI technologies to enhance the onboarding process for new hires within a corporate environment, leveraging large language models (LLMs) and retrieval-augmented generation (RAG) techniques. A comprehensive review of existing chatbot technologies was conducted, encompassing enterprise solutions, LLMs, and RAGs. The experimentation methodology examined various components of the RAG pipeline in three main stages. Initially, various preprocessing techniques were tested, including text augmentation with LLMs and structured data extraction from tables to enhance input data quality. Subsequently, the study explored different RAG configurations, ranging from straightforward single-shot models to more complex agentic models that incorporate iterative retrieval and reasoning. Finally, additional methods such as document tagging and filtering, as well as query expansion with a company-specific glossary, were explored to enhance the relevance of retrieved information, and ensure accurate interpretation of industry-specific terms. With a set of benchmark questions and reference answers tailored to the department's informational needs, the various approaches were then comprehensively evaluated using statistical metrics such as BERTScore, LLM-generated evaluations of accuracy, coherence, conciseness, and completeness, alongside efficiency metrics such as token usage, LLM calls, and response times. The results demonstrate that LLM-augmented text preprocessing enhances model performance, while single-shot RAG models provide the highest retrieval accuracy. Despite challenges related to information quality and AI hallucinations, the study highlights the potential of generative AI to streamline access to departmental information, thereby accelerating employee onboarding and improving workflow efficiency. Future research should focus on refining information quality and exploring more robust integrations of retrieval and generative techniques.

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Chapter 1

Introduction

1.1 Background

The department's Virtual Assistant was developed to enhance workflow efficiency and improve access to essential information. It features two main components: workflow automation, which streamlines routine tasks, and Q&A capabilities, which provide quick and accurate access to departmental knowledge. As the department grows and its processes become more complex, the Virtual Assistant's role has become increasingly critical. While the automation capabilities have been successful in improving workflow efficiency, there is an opportunity to further develop its Q&A capabilities to better support the department's evolving needs. This is particularly relevant during the onboarding of new employees, where providing accurate, contextually relevant information is key to helping them understand the department's structure, processes, and systems.

1.2 Problem Statement

As the department continues to expand, the limitations of the current Q&A capabilities within the Virtual Assistant have become more pronounced. Originally designed with rule-based mechanisms and a static knowledge base, the system effectively manages predefined FAQs and routine queries. However, it lacks the flexibility to provide accurate and contextually relevant responses to more complex or nuanced questions beyond the scope of its predefined knowledge base. This limitation is particularly evident during the onboarding process, where new employees need detailed, context-specific information to quickly understand the department's structure, processes, and systems.

To overcome these limitations, this research proposes the integration of generative AI, specifically large language models (LLMs), to expand the Virtual Assistant's capabilities. By leveraging LLMs, the system can offer more dynamic, context-aware responses, thereby improving the onboarding experience for new employees and streamlining access to critical departmental information for all users.

1.3 Research Objectives

The objectives of this research are hence to:

- **Explore Chatbot Infrastructure and Digital Tools:** Investigate the current landscape of digital tools and technologies, such as Microsoft Power Automate and Copilot Studio, and assess their suitability for integrating with generative AI to enhance workflow automation and information retrieval within a corporate setting.
- **Explore Techniques for Enhancing Question Answering:** Examine various document preprocessing methods, Retrieval-Augmented Generation (RAG) models, and other supplementary techniques such as document tagging and query expansion. This objective focuses on the practical application and refinement of these methods to improve the relevance and accuracy of information retrieval.
- **Evaluate Techniques and System Performance:** Assess the impact of implemented techniques on the performance of the generative AI system. This includes a comprehensive evaluation using statistical metrics like BERTScore, LLM evaluations of the quality of responses, as well as efficiency metrics.

1.4 Structure of the Thesis

This thesis is organized into six chapters. Chapter 2 provides a comprehensive review of the current Virtual Assistant, alongside an exploration of chatbot technologies, including enterprise solutions, Large Language Models (LLMs) and retrieval-augmented generation (RAG). Chapter 3 outlines the research methodology, detailing the experimentation scope and process, methods employed, and evaluation metrics used. Chapter 4 delves into the model experimentation process, exploring various preprocessing approaches, implementations of RAG, and other techniques. Chapter 5 summarizes the key research findings, offering a comparative analysis of the impact of the different preprocessing techniques, RAG models, and additional techniques on model performance. Finally, Chapter 6 concludes the thesis with a summary of the findings, an assessment of the study's limitations, and recommendations for future research.

Chapter 2

Literature Review

2.1 Review of Existing Chatbot

The Virtual Assistant plays a pivotal role in automating workflows and serving as a centralized information resource, with two primary capabilities: workflow automation, which automates routine tasks, and the Q&A functionality, which provides access to FAQs and departmental resources. A comprehensive review of the department's existing Virtual Assistant was conducted to assess its current performance, usage metrics, and areas for potential enhancement. While the system has been instrumental in enhancing operational efficiency, there are areas where further improvements could significantly increase its value to the department.

2.1.1 Underutilization of Q&A capability

Firstly, as the department grows and its processes become more complex, the underutilization of the Q&A capability suggests that it may not be fully meeting the department's needs.

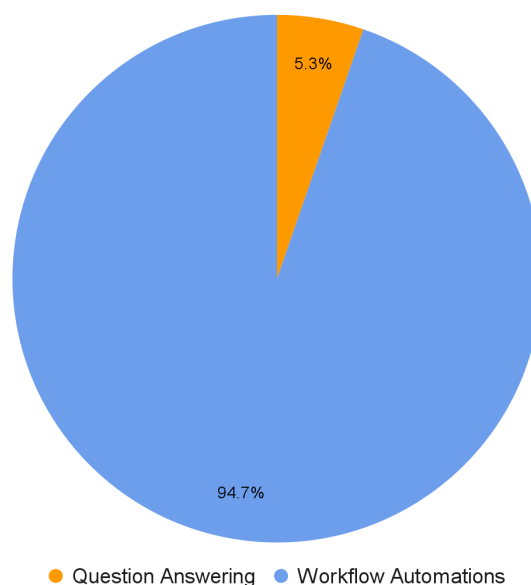


Figure 1: Comparison of Question Answering VS Workflow Automation Usage

Analysis of the past year's usage data reveals that 94.7% of interactions were related to workflow automation, while the Q&A component accounted for only 5.3% of interactions (Fig. 1). A line graph analysis over the same period confirms that this disparity is consistent over time (Fig. 2).

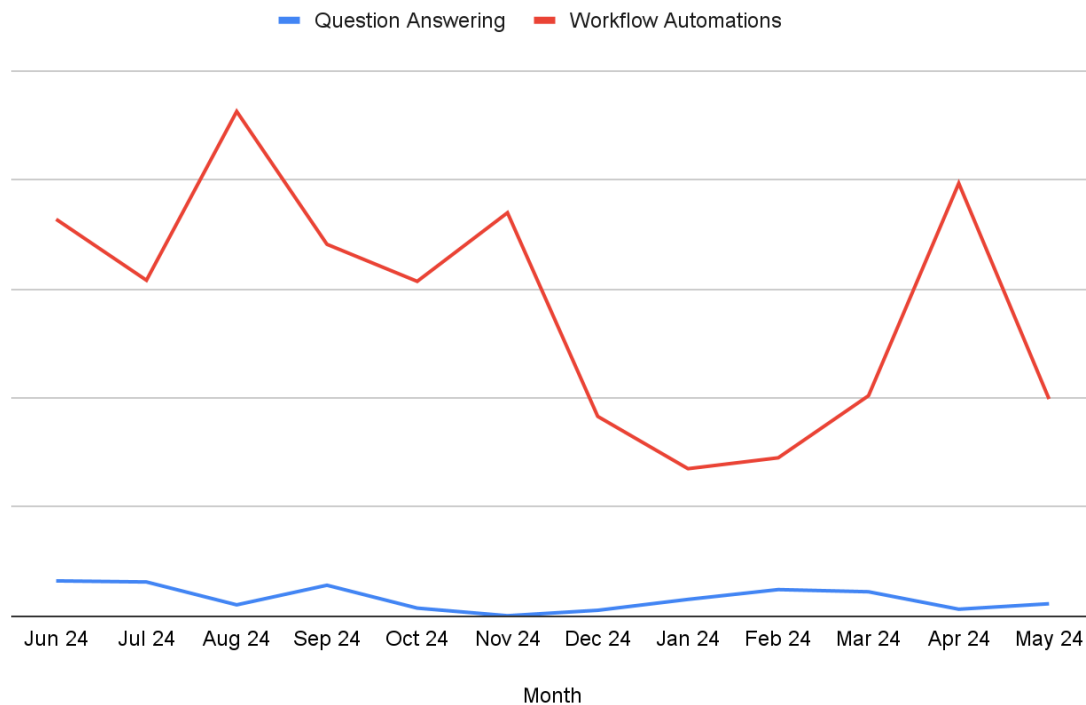


Figure 2: Comparison of Question Answering VS Workflow Automation Usage by Month

This usage pattern indicates that while workflow automation is well-aligned with departmental priorities, the Q&A capability could be enhanced to better support users seeking department information. The current system's reliance on predefined FAQs may limit its effectiveness in addressing queries that require dynamic information retrieval. There is considerable potential to transform the Q&A capability into a more dynamic, adaptive system that can provide timely and accurate responses, thereby increasing its value to users.

2.1.2 Quality and Relevance of Responses

The effectiveness of the Q&A capability is further constrained by its reliance on rule-based mechanisms and a static knowledge base. While this setup is adequate for addressing routine and straightforward queries, it lacks the sophistication needed to respond to more complex or contextually nuanced questions. The problem is particularly pronounced during the onboarding of new employees, who require comprehensive and context-specific information

to understand the department's operations and integrate smoothly. Although these issues do not critically impede the onboarding process, they represent opportunities for enhancing efficiency and user experience.

2.1.3 Scalability and Adaptability

Another limitation of the Q&A capability is its rule-based architecture, which poses challenges in terms of scalability and adaptability. As the department grows and its processes become more complex, the need for frequent manual updates to the system becomes increasingly burdensome. This reliance on static, manually updated content makes it difficult to keep the chatbot aligned with the department's evolving needs, resulting in a tool that may quickly become outdated or less relevant. A more scalable solution would involve an adaptive system capable of ingesting and processing information from a curated set of documents, thus minimizing the need for constant manual intervention and ensuring that the Virtual Assistant remains accurate and effective as the department evolves.

2.1.4 Proposed Solution: Leveraging Generative AI

To address these limitations, this research proposes the integration of generative AI technologies, specifically large language models (LLMs), to enhance the Virtual Assistant's natural language processing capabilities. LLMs can significantly improve the system's ability to analyze and respond to complex, context-sensitive queries by generating more precise and relevant answers based on a broader and continuously updated knowledge base. These advancements will enhance the system's ability to meet the evolving needs of the department, ensuring it continues to provide valuable support in both onboarding new employees and facilitating access to critical information for all users.

2.2 Technological Review

Building on the proposed integration of LLMs, this section of the literature review examines the broader technological landscape to identify the most effective tools and strategies for implementing the solution. A comparative analysis is provided of available digital tools, including enterprise solutions such as Microsoft Power Automate and Copilot Studio, alongside custom development approaches such as Retrieval-Augmented Generation (RAG). The goal is to evaluate the strengths and limitations of these options in enhancing the Virtual

Assistant's functionality in order to identify the most effective strategy for integrating these technologies to best support the department's evolving requirements.

2.2.1 Comparison of Enterprise Platforms and Custom Development

A comparison was conducted between Microsoft's enterprise tools and custom development options. Microsoft's tools, such as Power Automate and Copilot Studio, offer a robust low-code platform that enables rapid deployment with seamless integration into the department's existing infrastructure, including databases, Office applications, and collaboration platforms like Microsoft Teams and SharePoint [13]. These tools are particularly effective in quickly establishing foundational chatbot infrastructure, managing basic workflows, and ensuring secure user authentication through built-in access control mechanisms. Recent updates to Microsoft Copilot Studio have introduced enhanced AI capabilities, enabling the creation of chatbots that can respond with dynamic prompts, actions, and information from internal and external sources [18].

However, despite these advantages, there are some limitations to relying solely on Microsoft's tools. While the graphical interface simplifies development, it requires considerable manual effort to configure and maintain topics and conversation logic, which can become time-consuming as the chatbot scales. Furthermore, the platform's pre-built solutions, while user-friendly, offer limited customization options, particularly in fine-tuning information processing and data retrieval processes. Finally, due to the organization's data privacy policies, the full potential of Microsoft's generative AI features could not be leveraged, necessitating alternative solutions.

To address these issues, custom development was thus explored as a solution, particularly for handling the more complex backend tasks of the chatbot. A custom backend would allow for the development of advanced retrieval systems, such as a retrieval-augmented generation (RAG) architecture, which could enhance the chatbot's ability to extract relevant and accurate information from company documents and databases. Unlike Microsoft's out-of-the-box tools, custom development provides greater flexibility in both the design of the system and the ability to tailor it to the specific data retrieval needs of the department.

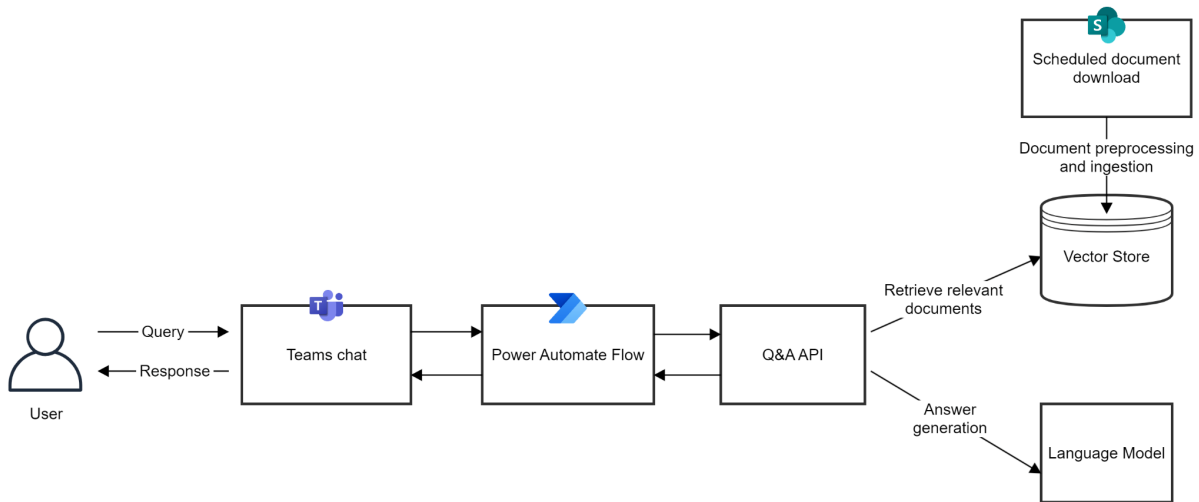


Figure 3: Hybrid Solution Design using Microsoft Enterprise Tools and Custom Development

Ultimately, the decision was made to adopt a hybrid approach (Fig. 3), combining Microsoft’s infrastructure for the chatbot's front end and integration with Office tools, with custom development for the backend. This allows the Virtual Assistant to benefit from both ease of deployment and advanced functionality. This blended solution optimizes the chatbot’s performance, ensuring it meets the department’s evolving needs.

2.2.2 Exploration of Large Language Models

Having established that custom development will be implemented, this section focuses on the exploration of Large Language Models (LLMs) and advanced techniques such as Retrieval-Augmented Generation (RAG) to enhance the performance and accuracy of the chatbot. LLMs have demonstrated exceptional capability in generating natural language, facilitating enhanced conversational experiences. Brown et al. (2020) highlighted that LLMs excel as few-shot learners capable of adapting to various language tasks with minimal training examples [1], while Kojima et al. (2022) explored their potential as zero-shot reasoners, discovering that these models can perform complex reasoning tasks without explicit prior training on the task format or domain [2]. This ability to handle diverse scenarios and downstream tasks makes them an attractive option for rapid development.

However, despite these advantages, one of the fundamental risks associated with LLMs is their tendency to produce hallucinations, generating plausible but factually incorrect statements. Ji et al. (2023) have highlighted that LLMs lack an inherent fact-checking

mechanism, leading to inconsistencies in outputs [3], while Xu et al. (2023) further assert that hallucination is an innate limitation of LLMs and cannot be fully eliminated [4]. A preliminary empirical exploration conducted with large language models (LLMs) such as OpenAI, Gemini, Mistral, and Cohere confirmed this (Appendix 1). A test case was designed where each model ingested a document and generated responses based on the content. It was found that, despite their impressive natural language capabilities, none of the models were immune to hallucinations. Such hallucinations thus pose a significant risk, especially when LLMs are used in professional environments where information accuracy is crucial.

To address the limitations of LLMs, the Retrieval-Augmented Generation (RAG) architecture has gained attention in recent research. Introduced by Lewis et al. (2020), it combines the generative abilities of LLMs with a retrieval layer, ensuring that responses are grounded in real-time, contextually relevant data retrieved from external sources, thus significantly improving response accuracy [5].

Recent advancements in Retrieval-Augmented Generation (RAG) technologies have been comprehensively surveyed by Gao et al. (2024) [6]. Beyond pre-retrieval optimizations such as index structure optimization and metadata integration, and post-retrieval enhancements such as re-ranking and context compression, the survey highlights the shift towards flexible architectures that incorporate specialized modules for tailored tasks. Further exploring the development of autonomous agents, Wang et al. (2024) [7] examine agents capable of self-directed planning and actions, detailing how early innovations such as ReAct synergize reasoning with action within LLMs to enhance decision-making processes [8].

This review of recent advancements in Large Language Models and Retrieval-Augmented Generation highlights key technologies that can be integrated into the experimental framework. This paper specifically implements and adapts agent models like PlanRAG by Lee et al. (2024), which integrates a preliminary planning phase before retrieval [9], and ReWOO by Xu et al. (2023), which aims to boost computational efficiency by decoupling reasoning from data gathering processes [10], aiming to directly address the challenges of hallucination and response accuracy.

Chapter 3

Research Methodology

This section outlines the methodologies employed to evaluate the effectiveness of integrating generative AI into the department's Virtual Assistant. The research process involved collecting user input, designing and testing various AI models, and assessing the performance of these models using both quantitative and qualitative metrics.

3.1 Experimentation Methodology

Several experiments were conducted to assess the effectiveness various techniques in data preprocessing, information retrieval, and answer generation. The key areas of experimentation included:

1. **Preprocessing Approaches:** Various preprocessing methods were applied to the input data to improve the quality and structure of the information.
2. **Models:** Several retrieval-augmented generation (RAG) techniques were explored, ranging from single-shot RAG to agentic models.
3. **Tagging and Filtering:** Document tagging and filtering methods were employed to improve the relevance of the retrieved information.
4. **Glossary Integration:** A glossary of departmental acronyms was incorporated into the query expansion process to enhance understanding of domain-specific language.

Each of these experiments was aimed at determining the most effective combination of preprocessing methods, model configurations, and information retrieval techniques to optimize the performance of the Virtual Assistant.

3.2 Development of Benchmark Questions and Reference Answers

To evaluate the effectiveness of each approach, a critical step involved developing a comprehensive and representative set of questions along with reference answers, which served as a benchmark for comparing the relative performance of the approaches.

Firstly, the questions were formulated through a series of interviews and consultations with multiple stakeholders, including end-users and management. The questions covered a wide range of topics, such as general department queries, roles and personnel, and the use of technology and tools, and varied in complexity, from straightforward factual queries to more nuanced questions. In total, 25 questions were created, ensuring a comprehensive exploration of the department’s structure, responsibilities, personnel, and tools. Table 1 provides a summary of the question types and examples.

Category	Complexity	Examples	Count
General department questions	Simple	- What does <department> do? - What are the clusters within <department>? - Who are the cluster heads in <department>?	3
	Complex	What are the <department> clusters, heads and business functions they cover?	1
Team responsibilities	Simple	- What is <team> responsible for? - What does <team> handle? - Which cluster is <team> under?	5
	Complex	- What is the difference between <team> and <team>? - What are the teams under <cluster> and what do they do?	2
Roles and Personnel	Simple	- Who can I go to for questions on <topic>? - Who can I speak with for <topic>? - Who heads <team>? - Who are the heads in <cluster> and what teams do they lead?	7
Cross-team collaboration	Simple	- How does <team> support <team>? - Who other departments does <team> interact with? - What external stakeholders does <team> interact with?	4
Technology and tools	Simple	- What are the systems used for <functions>? - Which parts of <department> is the <system> impacting? - What is the difference between <system> and <system>, and how do I decide which to use?	3
Total			25

Table 1: List of Sample Questions classified by Category and Complexity

Reference responses were then generated using large language models (LLMs), with preferred answers selected by stakeholders through a voting process. Manual fine-tuning and fact-checking were then performed using actual company resources to ensure the accuracy and

relevance of the answers. The final set of 25 reference answers served as the baseline for evaluating the performance of various models, approaches, and techniques.

3.3 Performance Evaluation Metrics

To evaluate the performance of the models, several metrics were used to compare the candidate answers generated by the AI against the reference answers. Recent literature highlights the evolving complexity of assessing Large Language Models (LLMs), advocating for multi-dimensional evaluation frameworks to capture their full capabilities [14] [15]. Among these, Desmond et al. (2024) highlights the inadequacies of traditional metrics like BLEU and ROUGE for tasks that demand creative or diverse outputs, advocating for LLM-based custom evaluators which can better reflect the multifaceted nature of generative outputs [16].

Building on these insights, our performance metrics extend beyond conventional statistical measures to include LLM-generated evaluations:

- **BERTScore:** BERTScore [12] evaluates the semantic similarity between generated answers and reference texts by comparing their embeddings. This approach offers a nuanced assessment of text similarity, addressing the limitations of BLEU and ROUGE which do not account for semantic coherence.
- **LLM-Generated Evaluations:** In addition to the statistical metrics, the candidate answers were evaluated on several qualitative dimensions, each assessed by LLM-based evaluation, on a scale of 1-5:
 - **Accuracy:** How factually correct the generated responses were.
 - **Coherence:** The logical flow and consistency of the answers.
 - **Conciseness:** Whether the responses provided the necessary information in a brief and clear manner.
 - **Completeness:** The degree to which the generated answers addressed all parts of the query.
- **Effectiveness:** Effectiveness was measured by the number of questions answered.
- **Resource Consumption and Efficiency:** In addition to accuracy and qualitative evaluations, resource efficiency was also considered in evaluating the preprocessing approaches. Metrics like token consumption, processing time, and the overall time taken to generate a response were considered.

These combined metrics provided a comprehensive understanding of how well each model performed, both in terms of producing accurate content and operating within practical resource constraints.

3.4 Analysis and Visualization Techniques

Various data visualization and analytical techniques were employed to analyze the performance metrics of the models. These techniques helped identify trends, patterns, and areas for improvement. The following methods were used:

- **Radar Chart:** Radar charts were employed to compare the relative performance of each model across the different LLM-generated metrics, such as accuracy, coherence, conciseness, and completeness. This visual representation allowed for a quick overview of which models performed well across multiple dimensions.
- **Box Plot:** Box plots were used to examine the median and distribution of performance scores across different models. This method was particularly useful for identifying outliers and understanding the distribution of scores within each model variation.
- **Bar Chart:** Bar charts were utilized to provide a straightforward comparison of model performance across key metrics, displaying the mean scores for each metric.
- **Scatter Plot:** Scatter plots were used to visualize the detailed distribution of scores for specific metrics. By plotting individual scores, this method allowed for a more detailed comparison of specific metrics.

These visualization techniques offered valuable insights into the relative strengths and weaknesses of each approach, guiding the selection of the most effective AI techniques to improve the Virtual Assistant's information retrieval and response capabilities.

Chapter 4

Model Experimentation

This chapter outlines the experimentation process conducted to evaluate different preprocessing techniques and model configurations for optimizing the Virtual Assistant's performance.

The core experimentation environment was designed with a modular architecture to enhance flexibility and reusability, allowing for rapid prototyping, testing of various techniques and configurations, and fine-tuning of the system's overall performance. It consists of two primary components: a document preprocessing pipeline and a question-answering service (Fig. 4). The experimentation involved iterative testing across different configurations within and overall system performance.

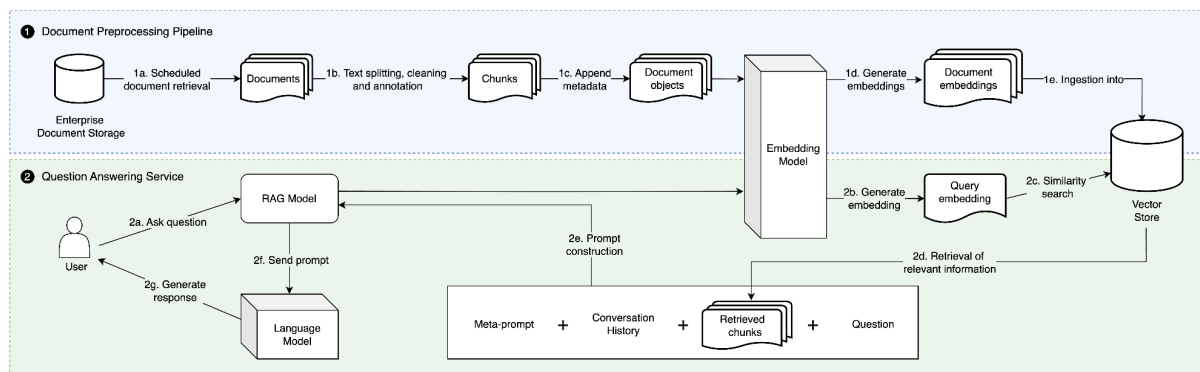


Figure 4: Overall Q&A Service Pipeline

The document preprocessing pipeline was designed with configurable parameters to facilitate a flexible approach to preprocessing techniques. Similarly, the base functionalities of RAG models were streamlined and standardized, including information retrieval, LLM interaction, and logging capabilities (Fig. 5). This enabled quick development and testing of agents, each adding specific functionalities while leveraging the same base components and ensuring consistency across all agents.

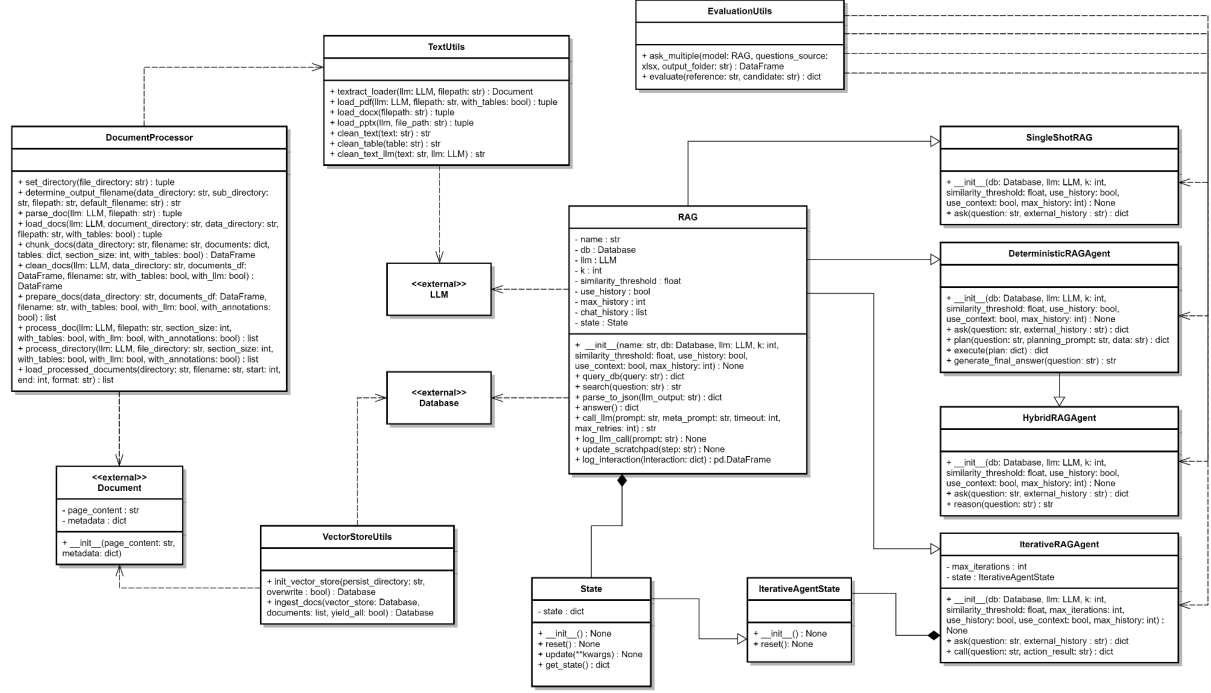


Figure 5: Modular Implementation of RAG Models

4.1 Preprocessing Approaches

A critical aspect of optimizing the Virtual Assistant's performance was the development and evaluation of various document preprocessing techniques. Given that the quality and structure of input data can significantly affect model performance, refining the preprocessing pipeline was a key focus of the experimentation process.

The primary data source consisted of presentation slides converted into PDFs, which posed unique challenges due to their structure and content. Each slide typically represented a distinct unit of information, but some, such as section dividers, contained minimal content. To maintain contextual integrity without introducing unnecessary complexity, a manual chunking method was adopted, grouping every three slides into a single section for preprocessing.

The content of the slides also presented significant challenges. Presentations often rely on visual elements and spoken narration, resulting in sparse text and fragmented paragraphs that are less suitable for processing by large language models (LLMs), which perform best with structured, coherent, and contextually rich text. The use of visual elements including charts, tables, and images add a layer of complexity to the interpretation of relationships between different pieces of text, risking the loss of critical information if not accurately captured. To

address this, AWS Textract was utilized to extract structured data, identifying tables within the documents [17]. These tables helped capture some of the spatial relationships between text elements, particularly for numerical and structured information. However, this approach had its limitations, as graphics such as flow charts or organisation charts did not adhere to strict formats, making them challenging to interpret accurately.

Four different preprocessing approaches were thus tested:

1. **Raw Text:** This approach used only the raw text extracted, without any restructuring or augmentation. This provided a baseline for evaluating the impact of more complex preprocessing methods.
2. **Raw Text with Tables:** This approach involved using the raw text along with any detected tables, which were structured in HTML format. The goal was to retain the table data and make it available for interpretation by the AI models, as tables often contained structured information.
3. **Cleaned Text:** In this approach, the text was augmented using a large language model (LLM) to improve the overall quality of the content.
4. **Cleaned Text with Tables:** This was an extension of the cleaned text approach, but with tables also incorporated into the data.

4.2 Retrieval-Augmented Generation (RAG) Models

With the preprocessing techniques in place to prepare the input data, the next phase of experimentation focused on optimizing the Virtual Assistant's ability to retrieve and generate accurate responses. This was achieved by evaluating various Retrieval-Augmented Generation (RAG) models, which integrated information retrieval with generative AI to enhance the system's performance.

Building on the modular framework, the agents were constructed following principles outlined in the ReWOO paradigm by Xu et al. (2023). ReWOO organizes reasoning, tool interaction, and synthesis into distinct modules: a Planner to create a sequence of interdependent steps, a Worker to retrieve external knowledge, and a Solver to synthesize all evidence for final output. This modular approach inspired the design of the Virtual Assistant's agents, ensuring flexibility and reducing redundant prompts. The agents in this experimentation were built to reflect similar modular principles, and encompass the following:

1. **Single-shot RAG:** A simple pipeline that performs a single retrieval and generates an immediate response.
2. **Iterative Agent:** A model that iterates through multiple retrievals, adjusting its plan dynamically based on the information retrieved.
3. **Deterministic Agent:** A systematic approach that creates and executes a predefined retrieval plan, ensuring all necessary information is gathered before responding.
4. **Hybrid Agent:** A flexible agent that either answers queries directly or generates a retrieval plan for more complex tasks.
5. **Two-Shot Agent:** A streamlined version that retrieves all required information in a two-step process, limiting the number of LLM calls to improve efficiency.

The **Single-shot RAG model** (Fig. 6) employs a straightforward approach, retrieving data in one pass from the vector store and using it to generate a response. Designed for efficiency, this model handles straightforward queries quickly by minimizing additional retrieval steps.

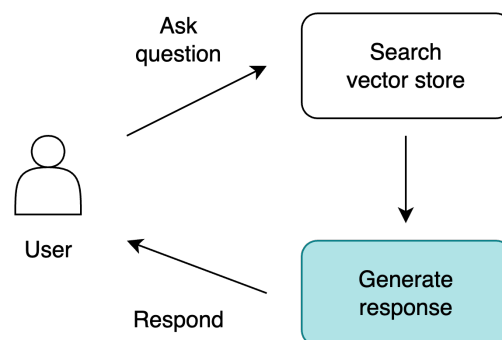


Figure 6: Single-shot Agent

The **Iterative Agent** (Fig. 7) addresses the limitations of single-shot retrieval by incorporating a flexible, multi-step process. It begins by creating an initial plan, then refines it dynamically as new data is retrieved, interacting with both the vector store and LLM. While its high autonomy and adaptability make it suitable for complex queries, this process can be resource-intensive, resulting in higher token usage and longer response times. Additionally, without proper safeguards, the agent risks entering infinite loops or compounding errors across iterations.

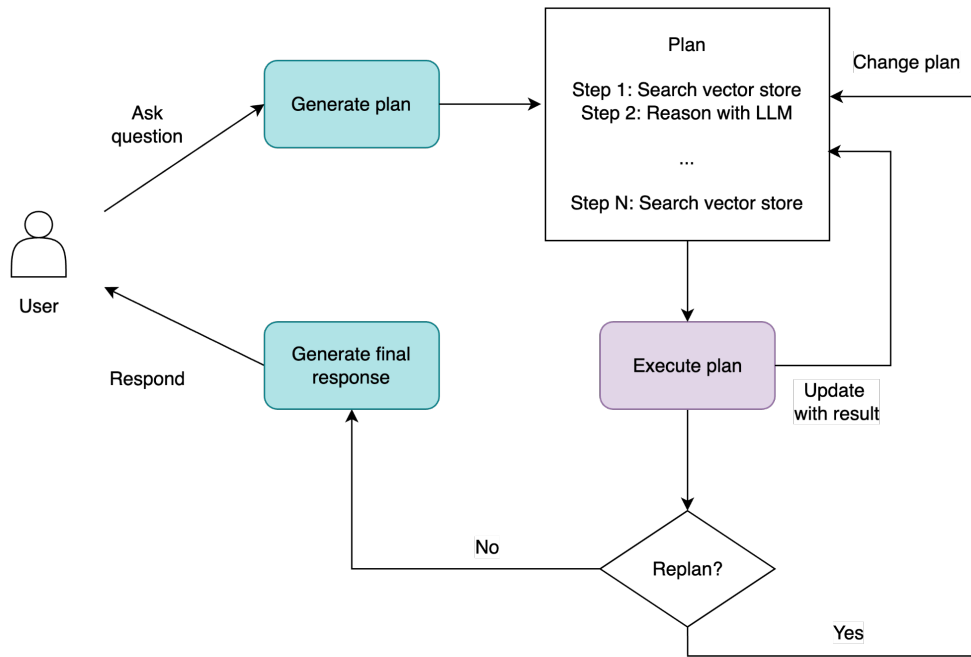


Figure 7: Iterative Agent

The **Deterministic Agent** (Fig. 8) offers a more methodical and predictable approach. It generates a fixed plan and sequentially executes the plan step-by-step before producing a final response. This method provides more predictable behavior compared to the Iterative Agent. While this approach is more efficient than iterative methods, as it minimizes overhead from multiple retrievals, it lacks the adaptability to adjust mid-process. As a result, the agent may be less effective in handling complex tasks that benefit from iterative refinement or dynamic adjustments.

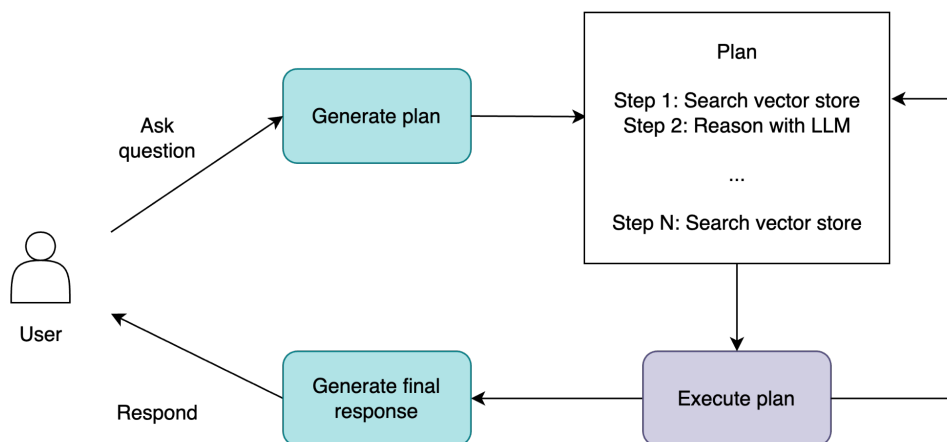


Figure 8: Deterministic Agent

The **Hybrid Agent** (Fig. 9) blends elements of both the deterministic and single-shot models. For simpler queries, the agent attempts to answer directly after the initial retrieval step. However, when faced with more complex questions, it generates and executes a plan, much like the deterministic agent. This hybrid approach strikes a balance between efficiency and adaptability, minimizing unnecessary retrievals while still being capable of handling intricate tasks requiring more detailed exploration.

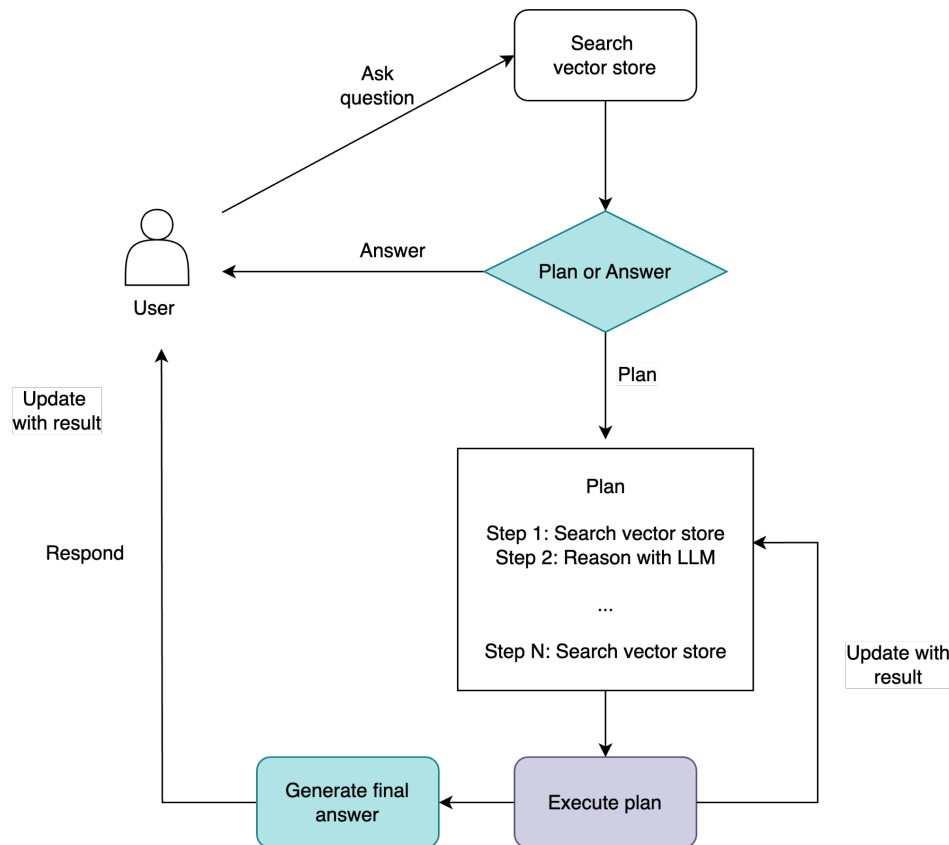


Figure 9: Hybrid Agent

Finally, the **Two-shot Agent** (Fig. 10) optimizes the hybrid approach by limiting the number calls to the language model. While it can perform multiple retrievals to gather relevant information, the agent does not engage in intermediate reasoning. In the first step, the agent retrieves all necessary data from the vector store and makes its initial LLM call to either generate a response or create a plan. If a plan is required, the agent assembles the relevant information and makes a final call to produce the output. This design balances efficiency and accuracy, reducing token usage while still ensuring all necessary information is gathered.

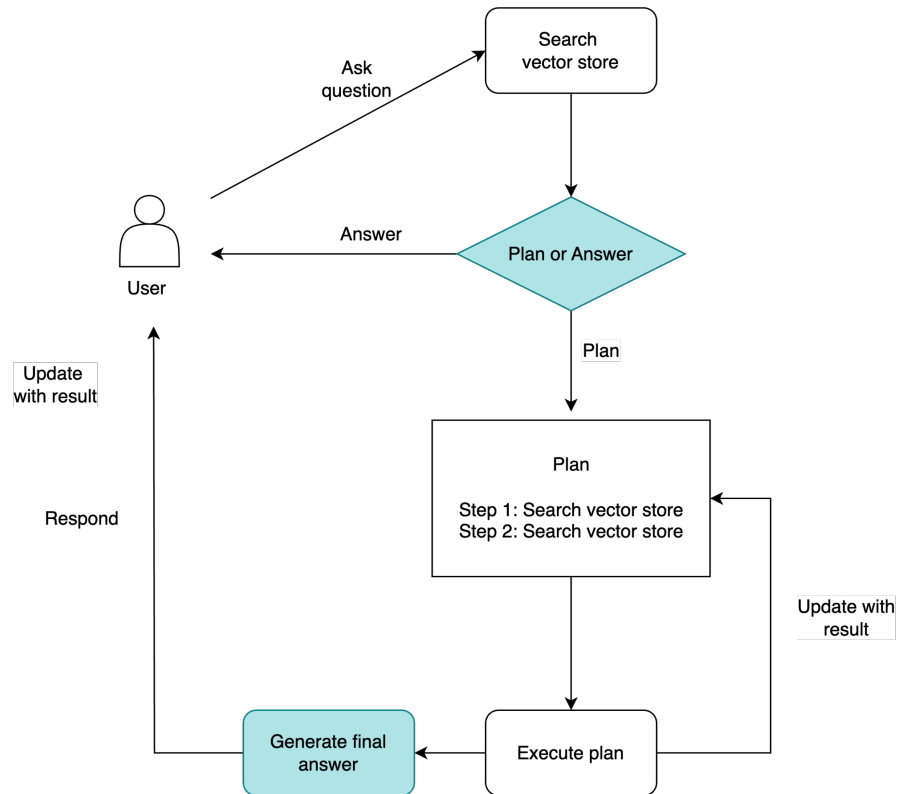


Figure 10: Two-shot Agent

4.3 Additional Techniques

In addition to testing preprocessing methods and RAG models, additional techniques were explored to further enhance model performance. For these experiments, the single-shot RAG model was used as a baseline to provide a consistent comparison point.

4.3.1 Document Tagging and Filtering

Document tagging was performed using an LLM to apply a predefined list of department-specific tags, ensuring accurate retrieval based on the internal structure of the department [11]. For query processing, keyword detection was handled by SpaCy's NLP capabilities. To filter documents based on the query, a similarity score boost was applied when document tags matched query terms, improving retrieval performance by prioritizing contextually relevant data without overly narrowing the document pool.

4.3.2 Query Expansion with Glossary

A glossary of company-specific acronyms and jargon was incorporated to enhance the search and retrieval process. This technique involved expanding user queries by adding relevant terms from the predefined glossary, which helped increase the similarity between the queries and documents containing these acronyms. By integrating these terms into the search process, the model was able to better interpret queries involving internal terminology, improving the accuracy of document retrieval in cases where company-specific language was crucial for understanding the query.

Chapter 5

Research Findings

This section provides a comprehensive evaluation of the preprocessing techniques employed in this study. Due to the non-deterministic nature of LLMs, each model was run five times on the same set of 25 questions to account for the variability in the model's outputs and to ensure a fair assessment of its performance. The performance of each approach was assessed based on multiple criteria, and the results are presented through various statistical analyses, including radar charts, box plots, and scatter plots, to provide a nuanced understanding of the impact of different preprocessing strategies on the model's output.

5.1 Evaluation of Preprocessing Approaches

This section delves into the comparative performance of various preprocessing techniques applied to enhance the quality of responses generated by the chatbot. Preprocessing plays a critical role in shaping how effectively large language models (LLMs) can interpret and respond to user queries. The evaluation focuses on different preprocessing strategies, including raw text, cleaned text, and the use of structured data formats like tables. Each approach was assessed based on key metrics such as accuracy, coherence, completeness, and efficiency. The goal was to determine which preprocessing methods resulted in the most accurate and contextually relevant outputs while balancing resource consumption and computational efficiency.

5.1.1 Performance Overview

In terms of overall performance, the cleaned text without tables approach emerged as the most effective preprocessing technique, as illustrated in the radar chart (Fig. 11). Table 2 shows a comprehensive comparison of the preprocessing approaches across all metrics. In terms of accuracy, this method achieved the highest accuracy score of 3.51/5 and the highest completeness score of 3.75/5. These results underscore the benefits of augmenting text with a large language model (LLM), which significantly improved the coherence and readability of the content. The enhanced clarity of the input data likely enabled the models to generate more accurate and contextually relevant responses.



Figure 11: Radar Chart of LLM-Evaluated Performance for Preprocessing Techniques

In contrast, the raw text approach, which involved minimal preprocessing, delivered the most coherent responses, scoring 4.43/5, and the most concise answers with a score of 3.91/5. The simplicity of the raw text, unaltered by additional processing, may have contributed to this outcome by allowing the model to focus on essential content without being influenced by potential noise introduced during text augmentation. This indicates a trade-off between the richness of content provided by cleaned text and the straightforwardness of raw text, each offering distinct advantages depending on the specific evaluation criteria.

Approach/Metric	Cleaned text only	Cleaned text with tables	Raw text only	Raw text with tables
Answered questions	115/125	109/125	120/125	115
Accuracy (BERT)	0.71	0.68	0.71	0.69
Accuracy (LLM)	3.51/5	3.35/5	3.43/5	3.38/5
Completeness (LLM)	3.75/5	3.50/5	3.65/5	3.48/5
Conciseness (LLM)	3.74/5	3.76/5	3.91/5	3.83/5
Coherence (LLM)	4.30/5	4.26/5	4.43/5	4.34/5
LLM Calls	1.00	1.00	1.00	1.00
Tokens	2126	3753	2069	3707
Time (s)	5.78	5.73	5.52	5.57

Table 2: Evaluation of Preprocessing Approaches Across All Metrics

5.1.2 Effectiveness in Query Resolution

When examining the number of questions answered correctly by each preprocessing approach, notable differences were observed (Fig. 12).

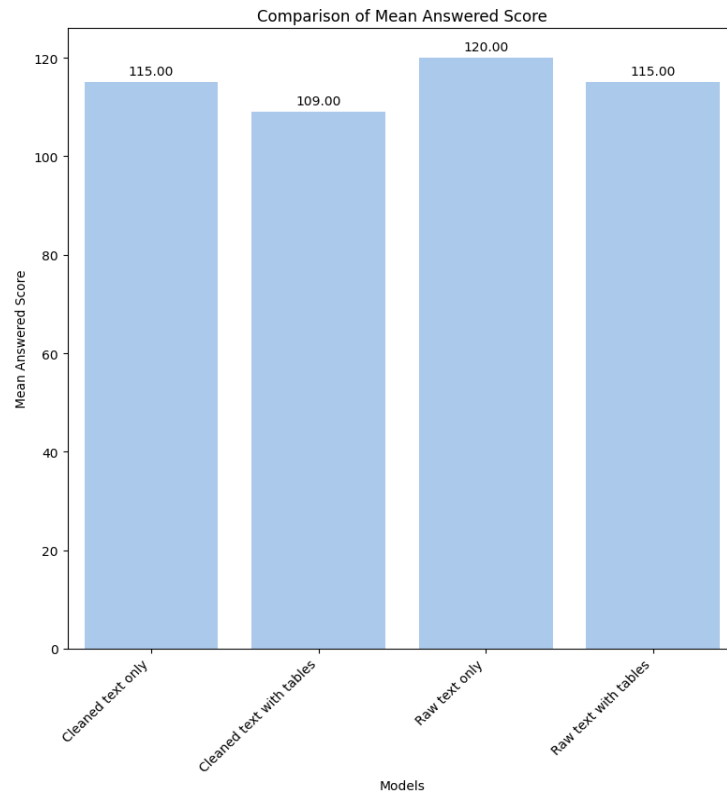


Figure 12: Comparison of Mean Answered Score for Preprocessing Approaches

The raw text approach without tables answered the highest number of questions (125 out of 125), indicating that minimal preprocessing allowed the model to interpret a broader range of queries effectively. This suggests that the raw text, despite its simplicity, provided sufficient context for most queries, avoiding potential overfitting or misinterpretation that might occur with more complex preprocessing.

The cleaned text approach without tables and the raw text with tables both answered 115 questions correctly. This similarity suggests that while cleaned text enhances content quality, and tables add structured information, neither approach drastically outperformed the other in terms of coverage. The slight reduction in the number of questions answered correctly compared to raw text only may be due to the introduction of processed or additional data, which in some cases could have added unnecessary complexity or redundancy, hindering the model's ability to generate correct responses.

Interestingly, the cleaned text with tables approach answered the fewest questions (109 out of 125). This outcome highlights the potential trade-offs involved in preprocessing: while this approach might produce more detailed responses in certain contexts (as discussed later in accuracy analyses), it also introduced variability and redundancy that likely hindered its effectiveness in answering a broader range of questions. The complexity added by both cleaning the text and integrating tables might have occasionally overwhelmed the model, reducing its ability to generalize across different queries.

5.1.2 Accuracy

In terms of accuracy, both BERTScore and LLM evaluations indicated that the cleaned text approach without tables slightly outperformed the other methods, though the differences were not substantial. BERTScore analysis showed that cleaned text achieved a mean accuracy score of 0.72, just marginally higher than the raw text approach, which scored 0.71 (Fig. 13). This small difference suggests that while text augmentation generally enhances accuracy, the raw text approach remains almost equally competitive when simplicity and directness are prioritized.

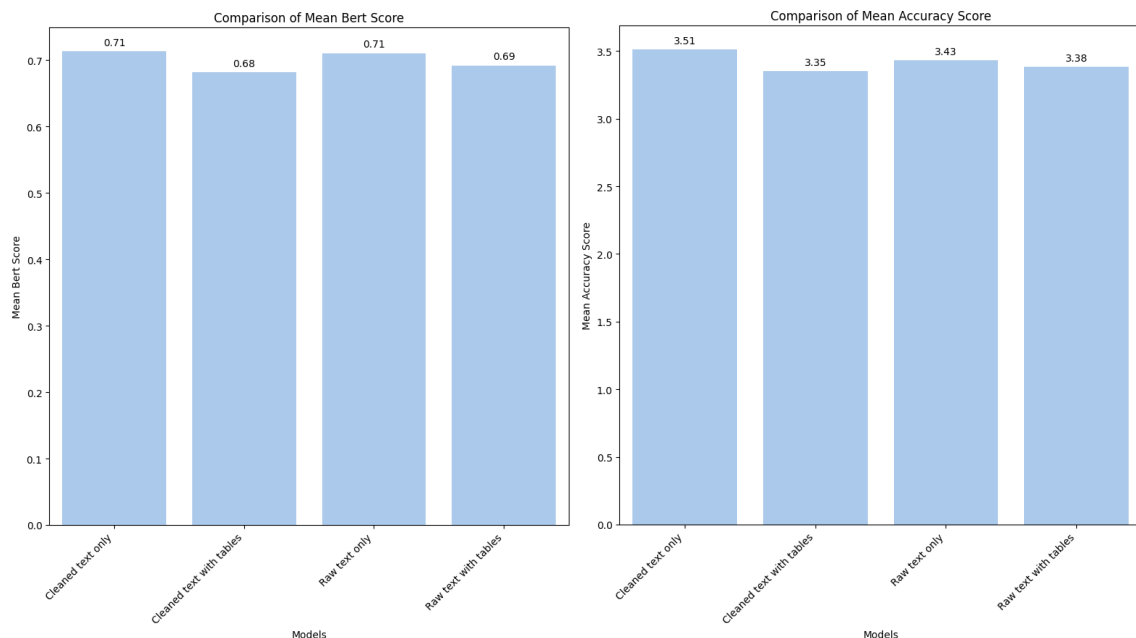


Figure 13: Comparison of Mean Accuracy (BERTScore and LLM-evaluated) for Preprocessing Approaches

Similarly, when evaluated by an LLM, the cleaned text approach obtained the highest accuracy score of 3.51/5, which is only slightly higher than the raw text score of 3.43, raw text with tables at 3.38, and cleaned text with tables at 3.35. These scores, although slightly varied, indicate that all preprocessing methods produced relatively comparable levels of accuracy, with no approach demonstrating a dramatically superior performance.

Interestingly, the cleaned text with tables approach, while yielding the lowest average accuracy score of 0.65, exhibited the highest interquartile range in accuracy. This means that although its median performance was lower, it also produced a wider spread of results, with some outputs achieving very high accuracy (Fig. 14). Similarly, the box plot for LLM-generated accuracy shows a similar pattern: the cleaned text with tables approach displayed the most variance but also included many of the highest accuracy scores among all approaches. This indicates that while the inclusion of tables introduces a degree of unpredictability, it can significantly enhance the accuracy of responses in specific cases.

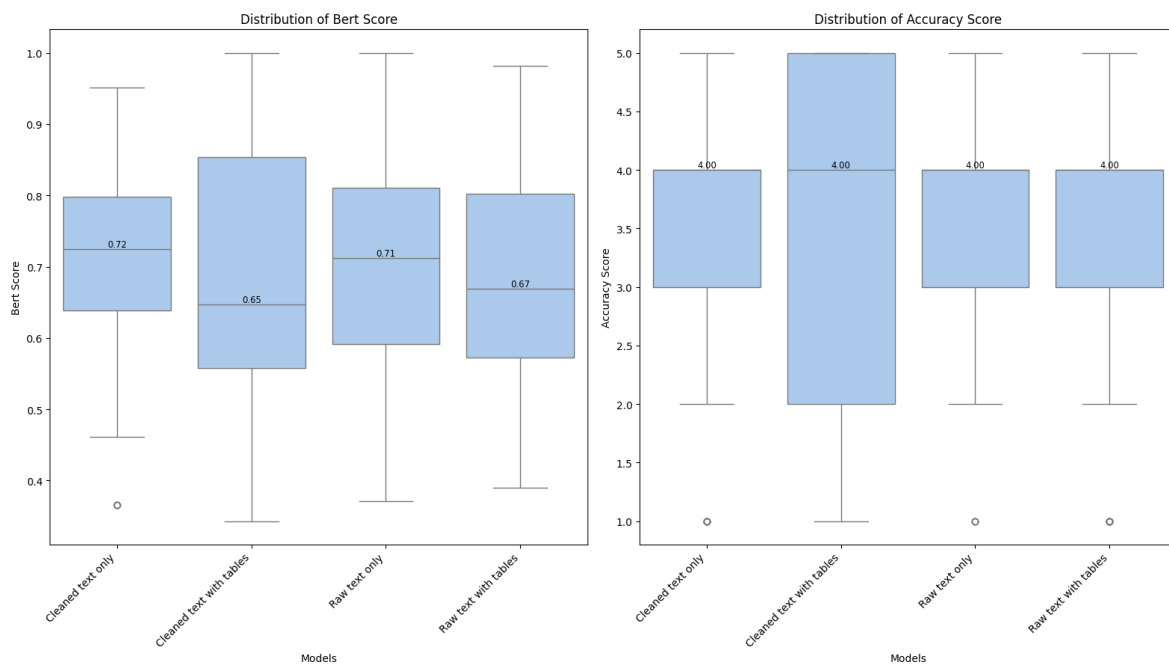


Figure 14: Distribution of Accuracy Score (BERTScore and LLM-generated) for Preprocessing Approaches

The scatter plot (Fig. 15) further illustrates this variability, showing that the cleaned text with tables had a broader distribution of accuracy scores, with more instances of both high and low accuracy compared to other approaches. This suggests that while tables can sometimes introduce redundancy and noise, potentially diminishing overall clarity, they also offer valuable

structured data that can improve responses for certain types of queries, particularly those involving complex information like organizational structure.

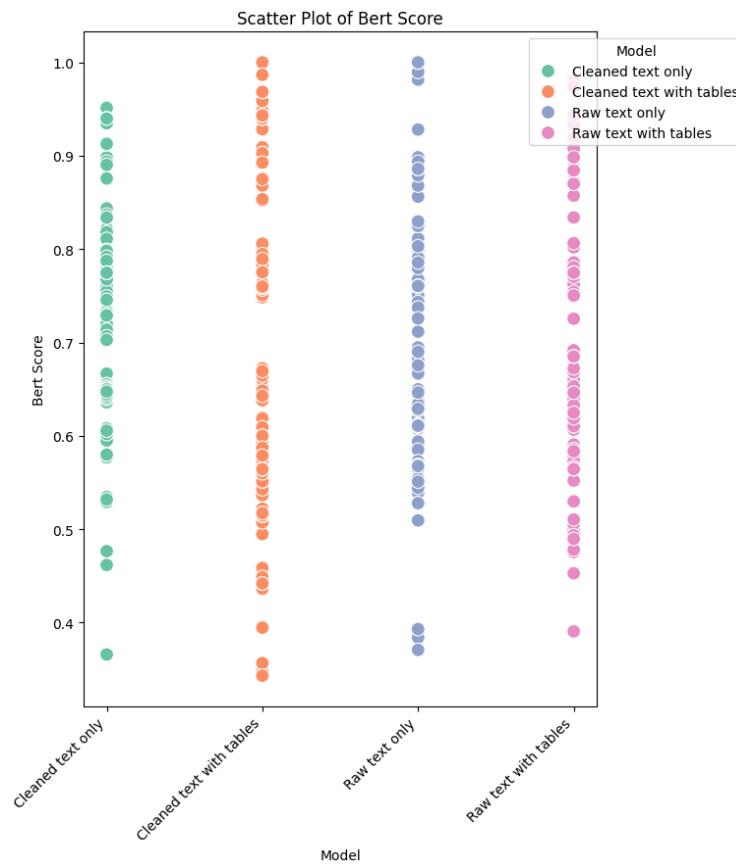


Figure 15: Scatterplot of Accuracy Score (BERTScore) for Preprocessing Approaches

Ultimately, these findings highlight the trade-offs involved in preprocessing decisions. While the cleaned text without tables provides more consistent and generally higher accuracy, the cleaned text with tables can offer superior results in specific contexts, albeit with greater variability. Given that these results are based on a limited set of 25 questions, the differences in accuracy might not be statistically significant enough to declare one approach categorically superior. Therefore, the choice between these preprocessing methods may ultimately depend on practical considerations, such as efficiency, computational cost, and the specific nature of the tasks at hand.

5.1.3 Resource Consumption and Efficiency

In addition to accuracy, efficiency was a critical factor in evaluating the preprocessing approaches, with particular focus on token consumption and processing time. The analysis,

depicted in the bar chart (Fig. 16), reveals that approaches incorporating tables were significantly more resource-intensive. Specifically, the cleaned text with tables approach consumed an average of 3,754 tokens per response, while the raw text with tables required 3,707 tokens. In contrast, the approaches without tables were markedly more efficient: the cleaned text consumed an average of 2,126 tokens, and the raw text 2,069 tokens.

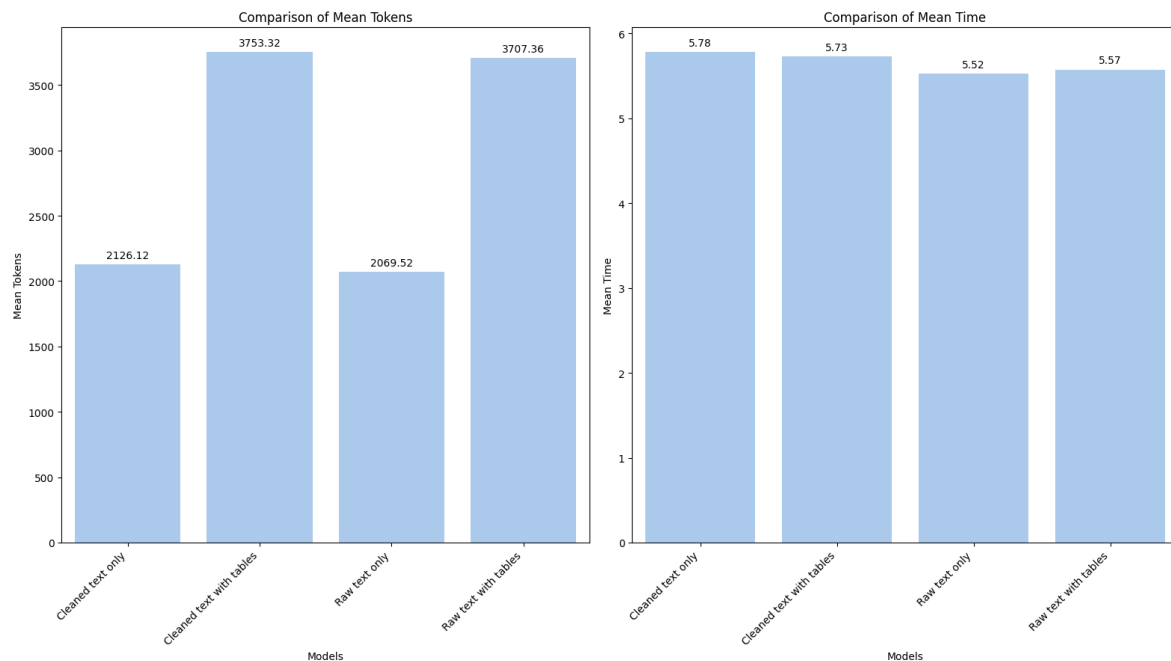


Figure 16: Comparison of Mean Tokens and Mean Time for Preprocessing Approaches

Despite the higher token consumption associated with table inclusion, this did not significantly impact processing time, with all approaches averaging around 5 seconds per response. This suggests that in scenarios where computational resources are constrained or where token efficiency is prioritized, the approaches without tables might be more suitable. However, in cases where the enhanced detail provided by tables is necessary, the slight increase in resource consumption might be justified.

This analysis highlights the importance of balancing accuracy with resource efficiency, especially in practical applications where both factors are critical to performance. The decision on which preprocessing method to use should therefore consider not only the nature of the data and the specific needs of the task but also the available computational resources and the cost implications of token usage.

5.1.4 Analysis and Interpretation of Results

In summary, while the cleaned text without tables emerged as the most consistent and accurate preprocessing approach overall, the inclusion of tables can be beneficial in specific scenarios, particularly for questions that require detailed numerical or structured data. However, the presence of tables also led to occasional redundancy, which may have diminished the clarity of responses in other contexts.

Given the limited scope of 25 questions, these differences in accuracy may not be statistically significant, suggesting that the choice between these preprocessing methods may ultimately depend on factors such as the specific application, resource availability, and cost considerations. For tasks that involve detailed numerical or structured data, the inclusion of tables might be justified despite the higher token consumption and variability. Conversely, for tasks where consistency and efficiency are prioritized, the cleaned text without tables may be the more suitable option.

Ultimately, these findings highlight the need for a context-sensitive approach to preprocessing. It is crucial to understand the document structure, characteristics, and techniques available to extract accurate information.

5.2 Evaluation of RAG Models

This section provides a comprehensive evaluation of several Retrieval-Augmented Generation (RAG) models by assessing their performance across four key dimensions: overall performance, accuracy, effectiveness in answering questions, and resource consumption and efficiency. The following sections offer a detailed analysis of the results, supported by diagrams, and examine the factors that contributed to the differences in model performance. The study's findings demonstrate the superiority of the single-shot Retrieval-Augmented Generation (RAG) model over agent-based approaches within this specific use case.

5.2.1 Performance Overview

The performance evaluation of the models highlights distinct trade-offs between accuracy, completeness, conciseness, and coherence, as evaluated by an LLM.

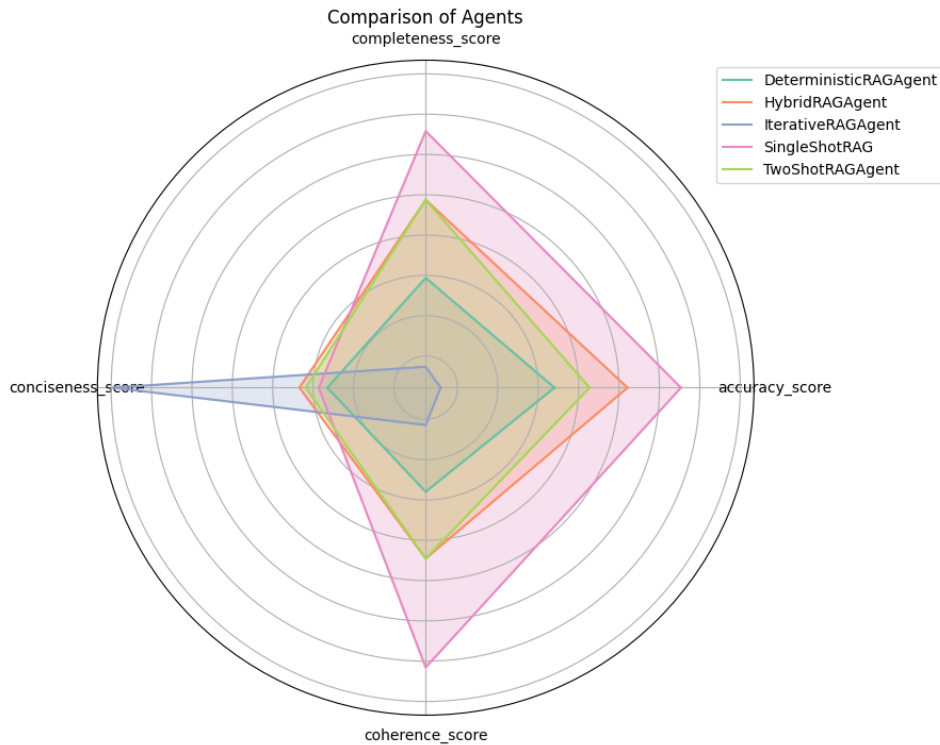


Figure 17: Radar Chart of LLM-Evaluated Metrics for RAG Model Comparison

In terms of overall performance illustrated in the radar chart (Figure 17) and Table 3, the Single-shot RAG model consistently outperforms other models in most metrics. It achieved the highest accuracy (3.64/5) and completeness (3.71/5), making it the most effective at generating responses aligned with the expected outputs. However, its conciseness (3.70/5) was lower than other models, indicating a tendency for longer responses.

In contrast, the Iterative Agent excelled in conciseness (4.46/5) due to its shorter outputs, but this came at the cost of accuracy (2.27/5) and completeness (2.31/5), with its iterative nature often leading to incomplete information retrieval.

The Deterministic Agent demonstrated moderate performance across all metrics, with accuracy (2.92/5), completeness (2.84/5), conciseness (3.66/5), and coherence (3.91/5). While not excelling in any single category, it provides a balanced approach, making it suitable for tasks requiring reasonable detail and brevity.

The Hybrid Agent achieved strong performance, particularly in accuracy (3.34/5) and coherence (4.06/5), striking a balance between generating accurate outputs and maintaining a concise, logical structure.

Similarly, the Two-shot Agent performed well in coherence (4.06/5) but lagged behind slightly in accuracy (3.12/5) and conciseness (3.74/5), producing slightly longer yet more structured responses.

Model	Single-shot RAG	Iterative Agent	Deterministic Agent	Hybrid Agent	Two-shot Agent
Answered	118/125	59/125	118/125	114/125	107/125
Accuracy (BERT)	0.71	0.47	0.64	0.67	0.64
Accuracy (LLM)	3.64/5	2.27/5	2.92/5	3.34/5	3.12/5
Completeness (LLM)	3.71/5	2.31/5	2.84/5	3.30/5	3.30/5
Conciseness (LLM)	3.70/5	4.46/5	3.66/5	3.77/5	3.74/5
Coherence (LLM)	4.31/5	3.76/5	3.91/5	4.06/5	4.06/5
LLM Calls	1.00	4.56	2.34	2.14	1.78
Tokens	1994.28	11749.31	4726.34	7407.49	6372.78
Time (s)	9.99	29.95	12.12	21.58	8.92

Table 3: Evaluation of RAG Models Across All Metrics

5.2.2 Effectiveness in Query Resolution

When examining the number of questions answered (Fig. 18), the single-shot RAG model and the deterministic plan-and-execute agent each successfully addressed 118 out of 125 questions, demonstrating their effectiveness in resolving queries.

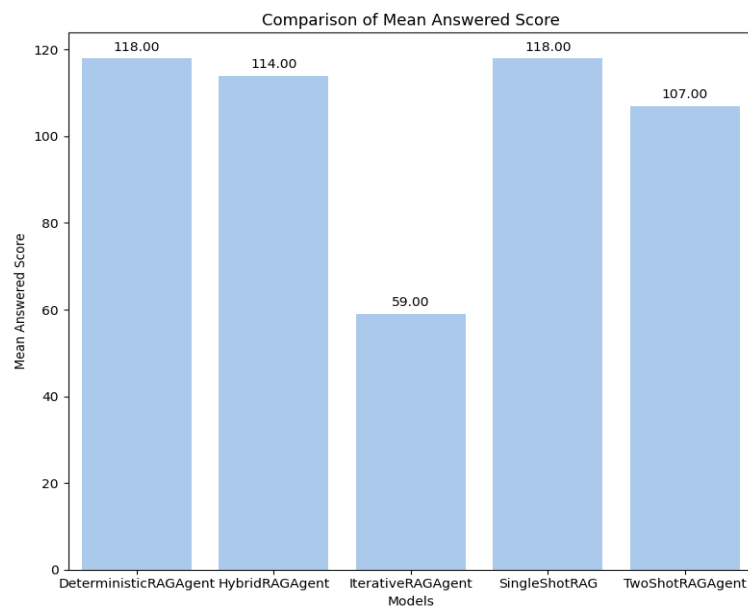


Figure 18: Comparison of Mean Answered Score for RAG Models

The hybrid agent followed closely with 114 questions answered, while the two-shot agent managed 107. In contrast, the iterative agent significantly lagged behind, answering only 59 questions, highlighting its limitations in scenarios where relevant information was either scarce or absent from the vector store.

5.2.3 Accuracy

In terms of accuracy, both LLM-evaluated accuracy and BERTScore were considered. Box plots (Fig. 19) demonstrate that the single-shot RAG model maintains a higher median accuracy (0.71) compared to other models, such as the hybrid agent and two-shot agent, both of which have a median BERTScore accuracy of 0.69. The distribution of scores indicates that lower accuracy scores are less frequent in the single-shot RAG model, emphasizing its reliability. Although the BERTScore differences between the single-shot and agent-based models were minimal, the LLM-evaluated accuracy shows a stark contrast: the single-shot RAG has a median score of 4/5, significantly higher than the median score of 3/5 observed in all other models except the iterative agent.

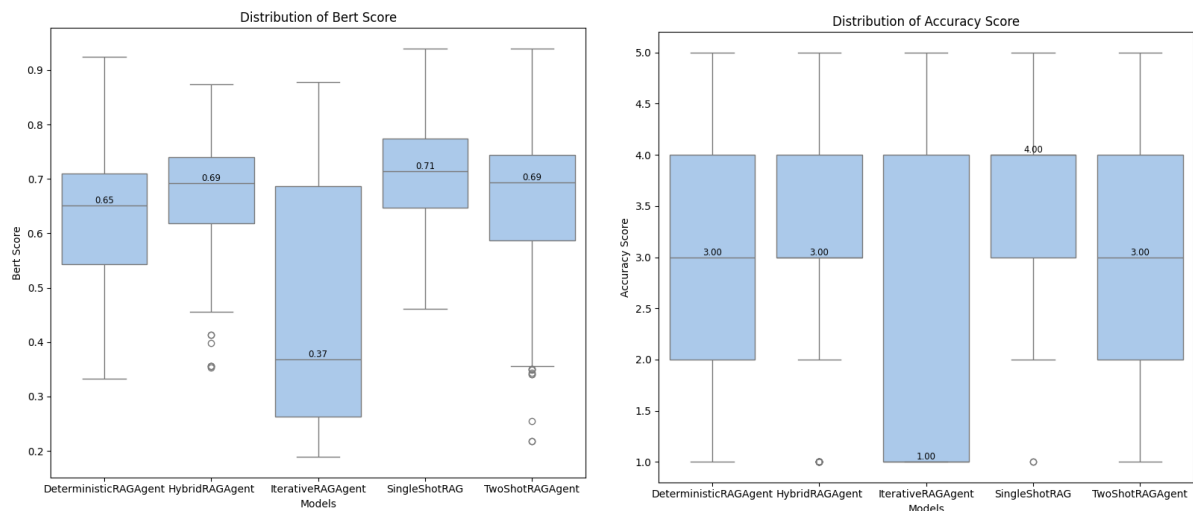


Figure 19: Box Plot of Accuracy Scores (BERTScore and LLM-generated) for RAG Models

5.2.4 Resource Consumption and Efficiency

Efficiency is a critical factor in evaluating the performance of these models, particularly in real-world applications where resources such as LLM calls, tokens, and time are limited. The study measured efficiency using metrics such as the number of LLM calls, the number of tokens consumed, and the time taken to generate responses (Fig. 20).

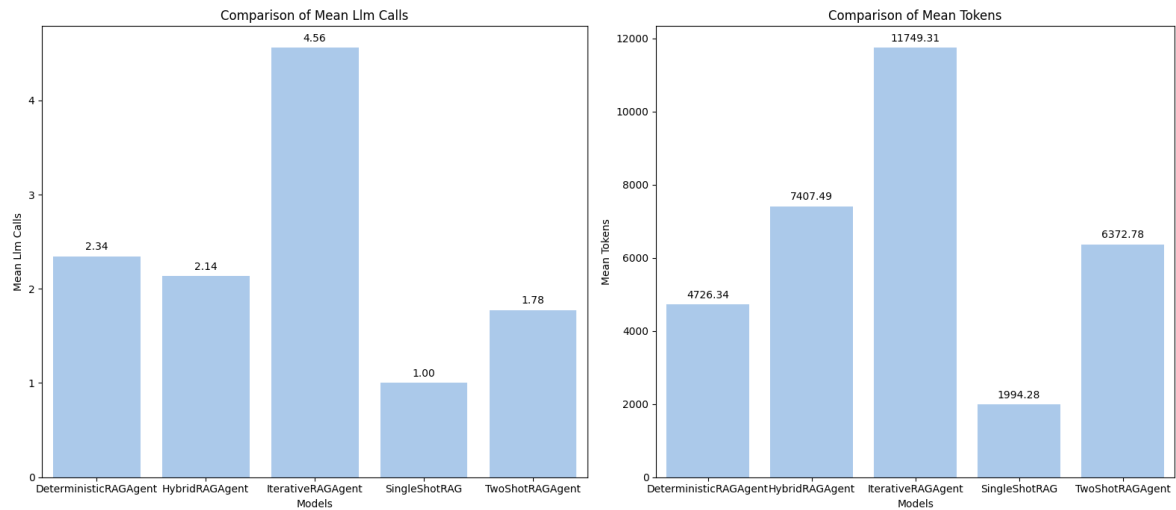


Figure 20: Bar Chart of Mean LLM Calls and Mean Tokens Consumed for RAG Models

The iterative agent was the least efficient, consuming an average of 4.56 LLM calls per query and a staggering 11,749 tokens, which is substantially higher than any other model. This high consumption is likely due to its iterative nature, where multiple retrievals and dynamic replanning require additional resources. In contrast, the single-shot RAG model, by design, required only one LLM call per query, consuming the least tokens at 1,994. The deterministic plan-and-execute agent, despite making more LLM calls on average (2.34), managed to consume fewer tokens (4,726) than the hybrid (7,407) and two-shot agents (6,372), indicating a more efficient use of resources in its systematic approach.

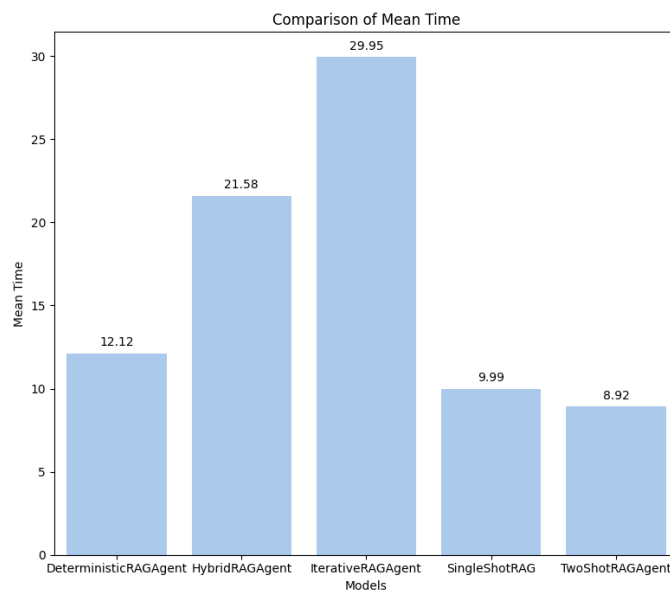


Figure 21: Bar Chart of Mean Time Consumed for RAG Models

When it comes to response time (Fig. 21), the iterative agent once again performed poorly, taking an average of 29.95 seconds to complete a query, the longest among all models. This extended time is attributed to the multiple steps involved in its operation, including iterative retrievals and replanning. The hybrid agent followed with an average time of 21.58 seconds, reflecting its complexity in balancing multiple retrievals and reasoning steps. The deterministic agent required 12.12 seconds on average, benefiting from its planned approach, while the single-shot RAG model took 9.99 seconds. Interestingly, the two-shot agent was the fastest, with an average time of 8.92 seconds, possibly due to variations in LLM API response times and its efficient two-step retrieval process.

5.2.4 Analysis and Interpretation of Results

The findings strongly suggest that the single-shot RAG model's straightforward approach is advantageous not only in accuracy and reliability but also in efficiency. The single retrieval step minimizes the resources required—both in terms of LLM calls and tokens—while still achieving high performance across other metrics. This efficiency, coupled with the model's ability to generate accurate and contextually relevant responses, makes it an ideal choice for scenarios where quick and reliable information retrieval is essential.

Agent-based models, while capable of performing multiple actions, introduced unpredictability due to their autonomous nature and reliance on complex prompt designs. Propagation of errors was a concern with agent models, as their iterative actions often compounded inaccuracies. Furthermore, multiple searches within a vector store were not beneficial when the relevant information was absent from the database, reinforcing the importance of information quality.

The Iterative Agent, while designed to refine answers through multiple retrievals, struggled when relevant data was missing or ambiguous. Its iterative process often led to futile retrieval attempts, compounding errors and reducing overall performance. The agent's flexibility was offset by inefficiency, as multiple LLM calls and higher token usage led to slower response times, making it less suitable for tasks requiring speed and precision.

The Deterministic Agent, though methodical and resource-efficient, was hindered by its rigid, predefined plans. This made it slower in responding to simpler queries where a more flexible approach like the Single-shot RAG would have sufficed. The Two-shot Agent

demonstrated better efficiency than the Iterative Agent, but its limited number of retrievals often resulted in incomplete data gathering, impacting accuracy and completeness.

The Two-shot agent, while more efficient than the iterative agent, still could not match the single-shot RAG's performance due to its inherent design, which involved limiting the number of retrievals. Although this approach reduced token usage, it sometimes led to incomplete data gathering, impacting the accuracy and completeness of the generated responses.

An important consideration in these findings is the limited skillset of the agents, which were restricted to using vector store search and reasoning. While the hybrid, deterministic, and iterative agents could employ reasoning, all models ultimately depended on the quality and completeness of information in the vector store. This limitation was especially detrimental to the iterative agent, which, despite its reasoning and dynamic replanning capabilities, could not overcome the absence of relevant data in the vector store. The hybrid and deterministic agents also faced similar challenges, though their more methodical approaches allowed for better performance in certain contexts.

Furthermore, given the nature of the questions in this study, which were mostly simple and required direct answers, the single-shot RAG model was particularly well-suited. Its straightforward approach matched the demands of the use case, allowing it to outperform more complex agents that were better designed for handling nuanced or multi-step queries. The lack of need for intricate reasoning or multi-step processing made the additional capabilities of the agent-based models largely unnecessary, further validating the single-shot model's efficiency and effectiveness.

In summary, the findings highlight that in scenarios requiring efficient information retrieval and reasoning, the single-shot RAG model's direct approach is superior. Conversely, the agent-based models, while offering greater flexibility and the potential for more comprehensive query handling, are more susceptible to error propagation and inefficiencies, particularly when faced with incomplete or poor-quality information sources.

Future work could explore hybrid models that combine the strengths of these approaches, potentially integrating the robustness of single-shot RAG with the flexibility of iterative agents, to enhance efficiency without compromising on performance in more complex information retrieval tasks.

5.3 Evaluation of Additional Techniques

This section delves into the evaluation of supplementary methods used to augment the capabilities of Retrieval-Augmented Generation systems. It specifically assesses the impacts of filtering techniques and query expansion through the use of a glossary.

5.3.1 Evaluation of Filtering Technique

The comparison between filtering and non-filtering approaches shows that while there were some minor differences in performance metrics, the overall impact was not significant (Table 4). Both methods answered a similar number of queries, with 115 for filtering and 117 for non-filtering, indicating that filtering did not substantially affect the system's ability to respond.

Approach/Metric	With filtering	Without filtering
Answered	115/125	117/125
Accuracy (BERT)	0.72	0.71
Accuracy (LLM)	3.56/5	3.49/5
Completeness (LLM)	3.79/5	3.73/5
Conciseness (LLM)	3.94/5	3.74/5
Coherence (LLM)	4.34/5	4.34/5
LLM Calls	1.00	1.00
Tokens	2378.84	2126.12
Time (s)	5.77	6.12

Table 4: Evaluation of Filtering Approach across all Metrics

In terms of accuracy, filtering resulted in only a slight improvement. BERT accuracy was 0.72 with filtering compared to 0.71 without it, and LLM accuracy saw a small increase from 3.49/5 to 3.56/5. While filtering helped align the retrieved documents more closely with the queries, the difference was minimal. Furthermore, although accuracy improved, the number of questions answered was higher without filtering (117/125 vs. 115/125).

Filtering did result in marginally better conciseness (3.94/5 vs. 3.74/5) and completeness (3.79/5 vs. 3.73/5). However, coherence remained unchanged between both approaches, with both scoring 4.34, showing that filtering had no noticeable effect on the logical structure of responses.

Efficiency metrics also demonstrated minimal differences. Both methods required the same number of LLM calls (1.00), and while token usage was slightly higher with filtering (2378.84 vs. 2126.12), the difference is not large enough to have a significant impact. The response time was slightly faster with filtering (5.77 seconds vs. 6.12 seconds), though this difference is relatively minor.

Overall, while filtering showed some small improvements in accuracy, completeness, and conciseness, the differences were not substantial enough to suggest a major impact. The results indicate that filtering had only a limited effect on performance, and the system's overall effectiveness remained similar regardless of whether filtering was applied.

5.3.2 Evaluation of Query Expansion with Glossary

The incorporation of a glossary of company-specific acronyms and jargon showed some effect on the document retrieval process, though its overall impact was modest (Table 5). The use of a glossary expanded user queries by adding relevant terms, increasing the likelihood of retrieving documents containing internal terminology. This approach aimed to improve the accuracy of results when queries involved company-specific language.

Approach/Metric	With glossary	Without glossary
Answered	124/125	118/125
Accuracy (BERT)	0.71	0.71
Accuracy (LLM)	3.58/5	3.50/5
Completeness (LLM)	3.70/5	3.72/5
Conciseness (LLM)	3.86/5	3.75/5
Coherence (LLM)	4.42/5	4.32/5
LLM Calls	1.00	1.00
Tokens	2030.84	2126.12
Time (s)	9.78	6.38

Table 5: Evaluation of Query Expansion with Glossary Approach across all Metrics

In terms of the number of queries answered, the system with the glossary answered considerably more queries (124 out of 125) compared to without the glossary (118/125). This suggests that incorporating the glossary allowed the model to better interpret certain queries, particularly those involving company-specific language.

However, the impact on accuracy was minimal. BERT accuracy remained identical at 0.71 for both approaches, and LLM accuracy saw only a slight improvement with the glossary (3.58/5 vs. 3.50/5). This indicates that while the glossary provided some improvement in understanding queries with internal terminology, it did not substantially increase the overall accuracy of the retrieval results.

Coherence similarly saw a minor improvement with the glossary (4.42 vs. 4.32), suggesting that the expanded query terms helped the model generate more logically consistent and contextually appropriate answers, particularly when dealing with company-specific jargon.

Interestingly, the glossary approach resulted in slightly lower token usage (2030 vs. 2126). However, the response time was longer with the glossary (9.78 seconds vs. 6.38 seconds), likely due to the additional processing required to expand the query with glossary terms.

Chapter 6

Conclusion

This thesis investigated the integration of generative AI technologies to enhance the onboarding process for new hires, emphasizing the utility of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques. Our findings confirm that LLM-augmented text preprocessing significantly enhances chatbot performance, with single-shot RAG models showing the highest retrieval accuracy. Additionally, while techniques like query expansion via glossary and document filtering were tested, their impact on improving performance was inconclusive. Overall, these findings indicate that such technologies can speed up the integration of new employees into corporate environments, thereby optimizing workflow efficiencies and reducing the burden on human resources.

The study also identifies key limitations and avenues for further development. One significant constraint was the reliance on a limited set of benchmark questions. Expanding the benchmark question set could help in refining the system's accuracy and adaptability in real-world scenarios. Additionally, the metrics for evaluation could be further developed by incorporating measures such as groundedness and reliability and refining evaluation prompts. Exploring reinforcement learning could also offer substantial benefits, allowing models to adapt and optimize their performance based on real-time feedback, thereby enhancing their applicability in dynamic scenarios.

The study has also highlighted the crucial role of information quality, particularly when processing documents like presentation slides. Future research can focus on exploring more robust methods for extracting and structuring information from visual-heavy content, leveraging advanced technologies such as GPT-Vision for processing visual data or using graph-based representations to structure and retrieve knowledge more effectively. These technologies could enhance the AI's understanding by providing richer, multi-dimensional data sources that reflect the complex interdependencies within organizational information.

Additionally, while this study tested a wide spectrum of RAG models—from single-shot to complex agent architectures—there remains considerable potential for refining each model type to maximize its effectiveness. Future research could systematically explore how

variations in prompts influence the AI agents' reasoning and output quality, refining prompt strategies to reduce error propagation, and optimizing prompts and tools to achieve more reliable and contextually appropriate responses.

In conclusion, while this thesis lays a substantial foundation for using generative AI to improve the corporate onboarding process, it also highlights the need for a broader experimental framework, enhanced data quality, and more sophisticated evaluation techniques. Future research should aim to build upon these findings, thereby fully realizing the potential of generative AI in transforming corporate environments.

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Appendix

LLM Model Comparison

Objective: A preliminary study was conducted with several AI models on Apple's 10Q Securities and Exchange Commission filing.

Prompt: Analyze Apple Inc.'s recent 10-Q SEC filing to summarize key financial data and operational insights. Provide an investment recommendation based on this analysis, addressing financial health, market position, and potential risks

Findings:

Model	Hallucinations	Incorrect information extraction	Faulty reasoning (e.g. inferring increase)	Wrong calculations	Recommendation
GPT-4	No	Yes	Yes, but less	N.A.	Hold for current investors; buy for new investors
Gemini	Yes	Yes, but less	Yes	Yes	Consider holding, buy on dips, invest for long-term
Mistral	No	Yes, but less	Yes, but less	N.A.	Apple is a strong investment opportunity
Cohere	No	Yes, but less	Yes, but less	Yes	Hold for long-term

Key observations:

- **Hallucinations:** Responses from all models were mostly grounded in observations from the data. However, the Gemini model inferred challenges and risks outside of the data.
- **Incorrect information extraction:** All models derived incorrect figures from the data, mixing up years or jumping between figures from three-months-ended and six-months-ended comparisons.
- **Faulty reasoning:** Cohere performed the best in terms of numerical reasoning, while other models had more inconsistencies, especially Gemini
- **Wrong calculations:** Of the models that attempted calculations (mostly percentage changes), none had correct calculations.